

How High-Quality TESTING Of MEDICAL DEVICES Is Ensured

Medical devices have a direct impact on the lives of patients. To deliver superior and safe experience to patients, medical device manufacturers are continuously innovating their devices using software. However, software come at the risk of defects and bugs

EFY BUREAU

Medical devices have a direct impact on the lives of patients and a single functional failure can mean the difference between life and death. To reduce unacceptable risk to patients, the medical industry is continuously innovating devices that enable better communication and performance. The World Health Organization has recommended to all governments worldwide to make national regulations for medical devices to provide assurance that devices will offer necessary risk reduc-

tion required to minimise harm in the event of a malfunction.

Medical devices' testing is required to detect the risks and effects of various environmental conditions. Safety assessment focuses on the reliability of products to function correctly and safely in response to inputs. It focuses on potential hazards from mechanical, electrical and environmental aspects of a design occurring during usage. Lifecycle, performance, compliance, interoperability and reliability testing are conducted to verify and validate

medical systems and their applications.

Medical device testing strategy

Medical devices are continuously regulated by multiple regulatory bodies for compliance to ensure their better performance and effectiveness. It is better to ensure compliance and conduct technical tests throughout the development cycle, starting from concept and design phase to production stage.

An effective medical device test

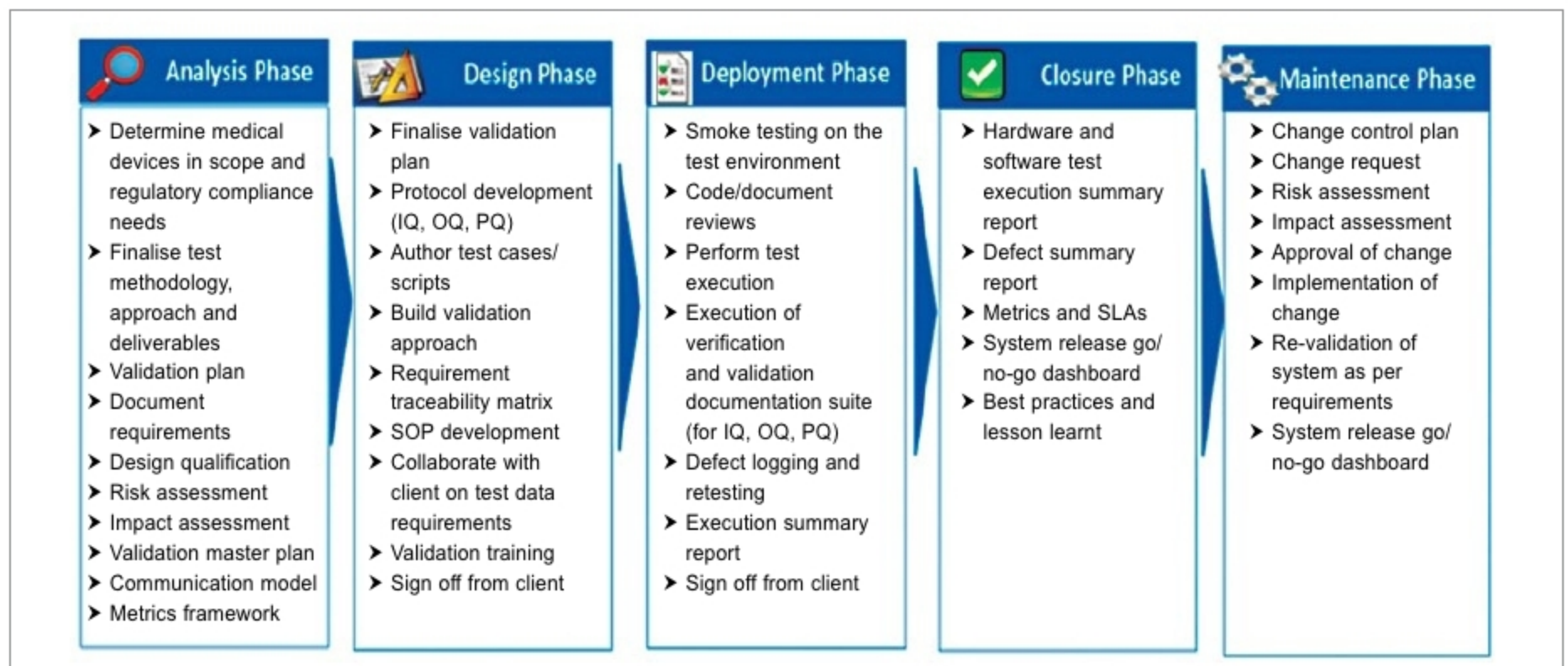


Fig. 1: Steps in medical device testing (Credit: Cigniti Technologies Ltd)